Sentiment Analysis of Code-Mixed Singlish (Sinhala-English) Reviews with Emoji Fusion NLP Approach

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I. Introduction

In today's Srilanka Almost every E-Commerce platforms and social media have seen a widespread adoption of "Singlish" a code-mixed combination pf Sinhala + English rendered in Roman Script [1]. Standard sentiment analysis tools designed for monolingual formal text perform very poorly on Singlish due to fluid intra-sentential Code-switching, informal spelling variations and High morphological complexity [2]. Recent natural language processing (NLP) research has increasingly explored multilingual pre trained transformer encoder for example **XLM-RoBERTa** for low resource Indian and South Asian languages [3, 4], User feedback online relies heavily on non verbal, visual affective markers specifically emojis. Emojis condense sentiment, emotional tone And sarcasm across language barriers sentiment, emotional tone, and sarcasm across language barriers [5, 6]. Recent multimodal frameworks demonstrate that fusing textual representations with dense emoji embeddings significantly enchances classification accuracy in codemixed scenarios [7]. In this project we check multimodel sentiment classification for Singlish product and application reviews by fusing contextual transformer representations with pre trained emoji embeddings.

II. Research problem

Now existing researches on the Singlish Sentiment analysis routinely discard emojis, emoticons, non alphanumeric symbols during text preprocessing, pinpointing them as irrecoverable noise or rather ignored [1, 4]. This raises Critical **research Gaps**:

1) **Loss of essential Affective Signals:** Emojis Carries vital emotional polarity and sarcasm in consumer reviews. For example Brief Singlish phrase paired with a fire 🔥 or with crying emoji 😭 Which is completely lost when the emojis are removed.
2) **Ommision of deep feature Fusion:** Current Singlish corpora rely mosttly exclusively in classical machine learning baselines (Logistic Regression, SVM) [1] or unadapted multilingual BERT without without cross-modal feature fusion [2]
3) **Data sourfce & Domain Limitations:** Existing Singlish Datasets are restricted to Youtube comments [1] leaving e-commerce product's feedbacks Like Daraz.lk, Social media marketplaces Like Facebook marketplace, Mobile app store reviews?.

Solving this problem is essential to enable accurate and automated market intelligence, product reputations tracking, customer experience analytics in Sri Lanka's digital economy.

III. Research Objectives

Primary Objective if this project is to develop, evaluate and release a benchmark emoji-fused sentiment analysis framework for CodeMixed singlish consumer reviews. The specific research goals are:

- **01 (Dataset Construction):** Harvest and annotate a new, gold standard Singlish review dataset from e-commerce sites, mobile app stores, facebook marketplace reviews that explicitly retains all inline Unicode emojis.
- **02 (Representational learning):** Extract contextual textual features by using fine tuned multilingual encoders (XLM-RoBERTa) and map them alongside 300-dimential emoji vector derived from `emoji2vec` and Unicode description specs
- **03 (Fusion Architecture):** Design and implement a Cross Modal Emoji Text Fusion Layer that eveluates Gated feature Fusion and cross attention modules to dynamically combine textual and visual affective signals.
- **04 (Empirical Evaluation):** Benchmarking the fused model against text only transformer baseline and classical ML classifiers across Accuracy, Macro F1 Score and per class sentiment metrics (Positive, Negative Neutral).

IV. Proposed method

Main Concept: The proposed solution uses a two stream deep learning network. The *Text Steam* Passes raw Singlish token sequences into a fine tuned XLM-RoBERTa encoder. The *Emoji Stream* extracts 300-dimensional continous representations for detected emojis usaing pre trained `emoji2vec` [6] and multi label emoji embeddings [7]. A Gated fusion model elarns dynamic attention weights to merge text and emoji representations into a unified feateure vector prior to softmax entiment classifications.

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